

CLINICAL PRACTICE GUIDELINES

CLINICAL STANDARD OPERATING PROCEDURE: INPATIENT DIABETES MANAGEMENT PROTOCOLS

*Evidence-Based Glycemic Control Frameworks, Critical Care Override Scales,
and Specialized Metabolic Care Pathway Integration*

This policy document establishes clinical protocol parameters governing glycemic monitoring, specialized basal-bolus correction insulin calculation models, emergency hypoglycemia intervention sequences, and perioperative metabolic guidelines at St. Jude General Hospital. Optimized for structural technical parsing and vector ingestion within the enterprise AI Knowledge Architecture.

Document ID: SOP-DIA-001

System Version: v3.2.0

Effective Date: July 1, 2026

Review Framework: 12-Month Standards
Cycle

Approval Authority: Endocrinology Directive
Board

Data Class: Clinical Knowledge Ingestion

1. Introduction, Objectives, and Scope

Inpatient hyperglycemia is linked to increased rates of surgical site infections, delayed wound healing, systemic organ failure, longer stays, and higher hospital mortality. This Standard Operating Procedure provides a clear framework for managing adults with pre-existing diabetes or hospital-induced hyperglycemia.

The goal is to maintain a stable, safe blood glucose target range of 140–180 mg/dL for most non-critically ill medical and surgical inpatients, and 140–180 mg/dL for critically ill patients in the ICU. Stringent protocols are required to minimize blood glucose fluctuations, avoid severe hypoglycemia (< 70 mg/dL), and standardize clinical handovers.

1.1 Target Audience and Enforcement

This policy applies to all attending physicians, mid-level providers, nursing teams, and clinical pharmacists across all adult inpatient departments, including general medical floors, surgical wards, and critical care units.

2. Clinical Accountability and Responsibility Matrix

Managing glycemic levels across multiple shifts requires coordinated, multidisciplinary execution with clear division of duties.

- **Attending Physician / Mid-Level Provider:** Responsible for ordering the baseline HbA1c testing within 24 hours of admission if no recent data exists, determining the specific glycemic control tier, and customizing basal-bolus or intravenous insulin orders.
- **Staff Nurse:** Accountable for point-of-care (POC) capillary blood glucose testing at exact scheduled windows, matching insulin delivery timing with tray distribution, and initiating emergency hypoglycemia rescue protocols.
- **Clinical Pharmacist:** Tasked with conducting medication reconciliation, evaluating drug-nutrient interactions (such as with parenteral nutrition or scheduled steroids), and validating initial sliding scale calculations against renal function metrics.

3. Scheduled Point-of-Care (POC) Blood Glucose Monitoring Pathways

The frequency of capillary testing must be aligned directly with the patient's nutritional intake status to prevent unexpected glycemic drops or spikes.

3.1 Nutritional Intake Timing Triggers

1. **Patients Consuming Scheduled Meals:** POC testing must be performed immediately prior to each meal (Breakfast, Lunch, Dinner) and at bedtime (22:00). Correction insulin doses must not be administered if meals are delayed or refused.
2. **Patients Under NPO (Nil Per Os) Status:** Capillary monitoring must transition to a strict every-6-hours loop (06:00, 12:00, 18:00, 24:00). Sliding scale correction rules apply to these specific time checks.
3. **Continuous Enteral / Parenteral Nutrition:** Testing must be completed on an ongoing every-4-to-6-hours cycle, with intermediate adjustments managed through uniform corrections.

4. Core Treatment Framework: Inpatient Basal-Bolus Insulin Delivery

Relying solely on reactive sliding-scale insulin setups is associated with high glycemic variability and is explicitly discouraged for long-term management on general medical wards. The preferred strategy utilizes a proactive tripartite basal-prandial-correction regimen.

Component A: Basal Insulin Formulation

Administer long-acting analog variants (e.g., Glargine or Detemir) at a consistent time daily to address baseline hepatic glucose production. Dose adjustments should be reviewed every 48 hours based on morning fasting values.

Component B: Prandial / Nutritional Insulin Formulation

Administer rapid-acting variants (e.g., Lispro or Aspart) immediately prior to meals to cover carbohydrates. If a patient shows unpredictable intake patterns, the prandial dose may be given immediately after the meal based on actual consumption.

Component C: Correctional Insulin Component

Reactive correction sliding scales are layered onto basal-bolus configurations to address unexpected glycemic spikes. Correction scales are categorized into Low, Moderate, or High intensity based on body weight and estimated insulin sensitivity indices.

5. Critical Care Settings: Continuous Intravenous Insulin Infusion (CIII)

In the Intensive Care Unit (ICU), severe hyperglycemia, diabetic ketoacidosis (DKA), and hyperosmolar hyperglycemic state (HHS) must be managed using continuous intravenous insulin infusions to ensure precise control.

5.1 Hourly Monitoring and Adjustment Protocols

All patients on a active CIII line must have their blood glucose levels evaluated on an hourly basis via arterial line or fingerstick draws. Adjustments to the infusion rate must follow the standardized institutional dynamic titration matrix, which matches current glucose levels and the rate of change over the preceding 60 minutes.

CRITICAL SAFETY EMERGENCY VALVE: If a patient's capillary or arterial glucose level drops below 70 mg/dL, the continuous intravenous infusion line must be paused immediately. The nurse must execute the emergency hypoglycemia algorithm and notify the critical care provider within 5 minutes.

6. Exceptional Situations and Hypoglycemia Rescue Protocols

Hypoglycemia (< 70 mg/dL) represents an immediate neurological threat and demands rapid intervention, bypassing non-essential administrative steps.

6.1 Step-by-Step Hypoglycemia Management (The Rule of 15)

1. **Identification:** Glucose reading < 70 mg/dL confirmed via immediate re-test.
2. **Intervention (Conscious Patient):** Administer 15–20 grams of rapid-acting oral carbohydrates (e.g., 4 ounces of fruit juice or oral glucose gel). Keep the patient NPO for solid foods.
3. **Intervention (Unconscious / Altered Patient):** Deliver 25 mL of 50% Dextrose (D50W) via secure intravenous line over 3 minutes, or administer 1 mg of Glucagon intramuscularly if IV access is unavailable.
4. **Re-evaluation Cycle:** Repeat capillary testing exactly 15 minutes post-intervention. If glucose remains below 70 mg/dL, repeat the treatment step and notify the attending physician immediately.

7. Quality Assurance Framework and AI Retrieval Anchors

To ensure ongoing compliance with institutional safety thresholds, the Quality Assurance Committee conducts monthly reviews of random charts, tracking documentation and clinical outcomes.

7.1 Core Performance Metric Thresholds

- **Hypoglycemia Incidence Limits:** Target rate of severe hypoglycemic events (< 40 mg/dL) limited to $< 0.1\%$ of total hospital days.
- **Timing Alignment:** Prandial insulin administration must occur within a 15-minute window of tray delivery across all wards.

8. Automated AI Knowledge Assistant Search Queries

To facilitate instant retrieval of specific sub-sections by floor clinicians using natural language search inputs, the following technical anchor codes are mapped directly into this document:

- Query: [SOP-DIA-HYPOGLYCEMIA] Pulls up the Rule of 15 rescue steps and D50W emergency delivery parameters.
- Query: [SOP-DIA-BASAL] Extracts target calculation models for basal-bolus configurations.
- Query: [SOP-DIA-ICU] Generates the dynamic hourly titration metrics for intravenous lines.